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BLOOD DISORDERS

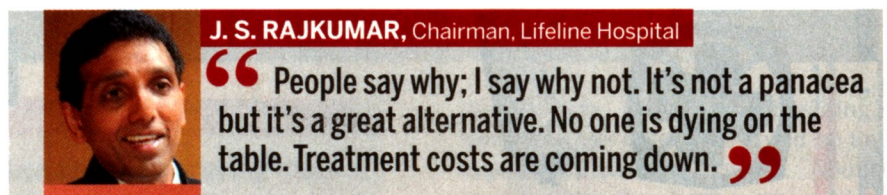
HARSHIL NANDA, 2

Harshil was diagnosed with thalassemia major at birth, a rare blood disorder where its ability to produce haemoglobin is defective. His parents, Ravin Nanda, a paan shop owner in Ahmedabad, and Trupti, were devastated. Stem cell treatment at the Gujarat Cancer Research Institute by Dr Sandip Shah was a success, making Harshil the second such patient to be successfully treated and showing a ray of hope to such patients in India.

STATUS IN INDIA

Treatment of disorders like thalassemia and cancers of the blood is only now gaining a foothold in India. While experts say the potential is good, it is far from being realised.

in earnest a decade ago. In Moscow, beauty salons began offering it as a cure for wrinkles, hair loss and dry skin but many of the patients reportedly ended up with tumours. In South Korea, a leading researcher was exposed for fraudulently claiming he had developed a human stem line through nuclear transfer. In India, P. Venugopal, a former AIIMS director, walked the ethical edge when he rammed through stem cell treatment for cardiac disorders when he was head of the institution. Since then, as Geeta Jotwani of ICMR points out, the DBT along with the ICMR have tightened controls and issued stringent guidelines for both research on stem cells and clinical trials. Balasubramanian says, "The problem is that it is more unethical than illegal, so you do have fly-by-night operators which patients have to be careful about." Bhan is for a legislation to be passed but prefers to wait for a year because, "so rapid



are the changes in stem cell research that the law may become outdated even before it is passed."

There is a clear need to tone down the hype about the benefits of stem cell therapy till solid basic research validates its efficacy. Jyotsna Dhawan, dean of the newly formed INSTEM, says, "We should not have stem cells hyped up as being a magic bullet. There is need for a lot more basic research to understand the mechanism that affects the degenerative and regenerative process. We need to understand how these cells are set aside and retained as a reserve population and remain neither divided nor differen-

tiated till the appropriate time. Also, how they go about repairing tissue." There are still huge gaps in the understanding of how the stem cells actually go about doing their work and how they will behave in the body. Doctors don't want to be putting a stem cell into the brain and discovering that it was a bone! Moreover as Sujata Mohanty, head of AIIMS Stem Cells Facility, points out, "There is a huge difference between what we achieve in the lab and its efficacy with patients. Each patient has his or her own recovery pattern and would need different cell dosages. So we still have plenty to understand about the processes." Getting stem cells to grow a tooth or as a cure for baldness is still a long way. Yet as Balasubramanian says, "Stem cells are both exciting and promising, and the future would only see an increase in this therapy." Therein lies the hope. *with inputs from*

Senthil Kumar and Uday Mahurkar

JYOTSNA DHAWAN, Dean, INSTEM

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